Team Poster Project

CSC3107: Information Visualization

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Published visualizations often influence how people understand public issues, scientific findings, business trends, or social problems. In this project, your team will act as an independent data visualization consulting team. You will select a published visualization, assess its strengths and weaknesses, reconstruct or adapt the underlying data, and produce an improved version for a specific audience and use case. Your goal is not to make the visualization more persuasive at any cost, but to deliver an evidence-based revision that is accurate, interpretable, transparent, and useful to its intended audience.

Teams of 5—7 students will work on this project across the trimester, culminating in a poster presentation during the Week 13 lab session. Throughout the project, teams are required to use one or more generative AI tools (such as ChatGPT, Claude, or GitHub Copilot) in a deliberate and documented manner, and to reflect critically on how those tools shaped their work.

1 Learning Objectives

- Demonstrate mastery of the technical skills acquired throughout the course.
- Manage a data visualization project from initial concept to final presentation.
- Evaluate and critique data visualization.
- Choose appropriate tools and techniques, demonstrating an understanding of available resources.
- Apply critical and creative thinking when analyzing and visualizing data.
- Document data and code to ensure clarity and reproducibility.
- Present information professionally as a poster, in both written and oral form.
- Write a design brief that identifies a target audience, an analytical task, and the design rationale linking design choices to that task.
- Engage with generative AI tools critically and responsibly, documenting how their suggestions were evaluated, adapted, accepted, or rejected.
- Collaborate effectively in a team.

2 Tasks

The project follows the standard arc of a visualization redesign: diagnose the original through critique, write a brief that anchors the redesign, gather and prepare the underlying data, implement the redesign, and present the result. Documentation of AI use runs alongside these tasks throughout.

1. **Critical Analysis:**

   a. Choose a published, thought-provoking example of data visualization from news media.
   b. Ensure that the chosen visualization is sufficiently complex to allow ample opportunities for applying the skills learned in the course.
   c. Critically analyze the strengths and weaknesses of the selected visualization.
   d. Discuss your findings and seek feedback from both your classmates and the instructors.

2. **Design Brief:**

   Before building the redesigned visualization, write a short design brief that anchors all subsequent design decisions. Your team is responsible for formulating this brief. It must identify a specific audience for the redesigned visualization (for example, readers of the original news article, visitors to a museum exhibit, or school science teachers), the analytical task or decision the visualization should support for that audience, and the design choices that serve that task.

   On the poster, the brief appears as a single, compact, prominently placed box of roughly 50–125 words. The brief is most effective when it links specific design choices explicitly to the task. For example:

   > **Design brief**
   >
   > This redesign is intended for a review article in an academic social science journal. The goal is to compare how housing affordability has changed across income groups.
   >
   > *Main revisions*:
   >
   > - Small multiples to unclutter the visualization
   > - Common vertical scale to facilitate comparison between groups
   > - Direct labels to avoid gaze switching between plot and legend
   > - Annotations to highlight the most important trends

   You may refine the brief as your understanding of the data and the redesign develops, but you should have a working version before you commit to a visualization design.

3. **Data Preparation:**

   a. Gather relevant data to support your visualization project.
   b. Document the data sources and describe all actions taken for data cleaning and transformation.
   c. You may choose to replace the dataset used in the original article with another related dataset, but you must demonstrate the relevance of the data to your chosen visualization. Do not present surrogate data as if they were the original article's data.
   d. Document the computer code used for data cleaning and transformation to ensure reproducibility.

4. **Data Visualization:**

a. Beyond merely reconstructing the original visualization, creatively apply the techniques learned in the course to improve it.
b. The design choices should demonstrably implement the design brief from Task 2: the visualization should serve the audience and analytical task you identified.
c. The outcome should communicate the data accurately with a visual style that supports readability.
d. Apply a grammar-of-graphics approach to construct the visualization.
e. Document the code used to produce the improved plot.

5. **Poster Presentation:**

   Create a poster that includes the following elements:

   - Introduce and display the original visualization, crediting its source.
   - Include a critique of the original visualization, suggesting improvements.
   - Present the improved visualization.
   - Describe the data used for constructing the improved visualization and state their source.
   - Include the design brief (see Task 2) as a compact, prominently placed box.
   - Provide references. If space is limited, you may include a QR code linking to a webpage with the full bibliography and any supplementary materials.

   In the final lab of the trimester, each team will be given 10 minutes to present their poster orally, with all members taking turns. During the presentation, the team should articulate the specific components of the design brief. The presentation is followed by a 5-minute question-and-answer session. Teams both give and receive feedback during the session; providing thoughtful feedback is part of the assessment.

6. **AI-Assisted Workflow:**

   Throughout the project, you are required to use one or more generative AI tools (e.g., ChatGPT, Claude, GitHub Copilot) in a deliberate and documented manner. Realistic uses include critiquing the original visualization, debugging R or Quarto code, suggesting `ggplot2` techniques, drafting poster text, or exploring alternative design options. You are expected to evaluate AI output critically: accept it when it is correct and helpful, refine it when it is partly useful, and reject it when it is wrong or misleading. The accompanying deliverables (an AI Interaction Record and an AI Usage Report) are described in Section 3.

3 AI Usage and Documentation

The use of and reflection on generative AI is a required and assessed part of this project. You must submit two AI-related deliverables in addition to the visualization itself.

3.1 AI Interaction Record

For each chat-based AI tool you used (ChatGPT, Claude, Gemini, and similar), submit the raw, unedited transcripts of the conversations you reference in your AI Usage Report. Most such tools let you export a conversation as text or Markdown; that export is exactly what we need. Do not edit, reformat, summarize, or otherwise tidy up the transcripts. The file name should identify the tool and, if known, its version (for example, `ai-log-chatgpt-5.md` or `ai-log-claude-opus-4.md`); submit as many or as few files per tool as needed to cover the conversations you cite.

You do not need to include every conversation. Include enough material to substantiate the specific claims you make in your AI Usage Report. If your Report says, for example, that the AI suggested faceting by income group rather than by year and that this improved comparability, the corresponding conversation must appear in the Record so that we can read the original exchange. Screenshots are not acceptable because they cannot be searched or copied.

The Record itself is not annotated and not graded for completeness or structure. Its purpose is to let us verify the claims you make in the Report: the reflective work belongs in the Report, and the Record is evidence. Claims in the Report that cannot be traced to the Record may not receive credit.

Tools that operate through inline code completions (e.g., GitHub Copilot) do not produce an exportable transcript. For these tools, no Record file is required; instead, support any claims you make about them by quoting the relevant prompt comments and surrounding code in the AI Usage Report itself.

3.2 AI Usage Report

The AI Usage Report is a short reflective report (≤ 2,000 words) explaining how generative AI shaped your work. Where you make a specific claim about an AI suggestion (for example, that it influenced a design decision, or that you rejected it for a particular reason), point to the corresponding exchange in the AI Interaction Record so that the claim can be verified. The report should cover the following points:

- **Prompting strategies.** How did you frame prompts? Did you iterate on them? Did you provide context (data, code, prior plot drafts)?
- **Influence on design decisions.** Identify at least two specific decisions (visualization, code, or poster) where AI input changed what your team would otherwise have done. Explain the change and why you judged it an improvement.
- **Corrections and rejections.** Identify at least two cases in which you corrected, refined, or rejected an AI suggestion. Explain what was wrong with the AI output and how you handled it.
- **Ethical and accountable use.** Briefly address verification of any factual or technical claims, attribution of any AI-generated phrasing, and limitations you observed (e.g., outdated information, plausible but wrong code).

If your team used more than one AI tool, the report should specify which tool contributed where; you do not need to write a separate report per tool.

The report is graded on the quality of reflection and the evidence of critical engagement, not on the volume of AI usage. A team that used AI sparingly but documented thoughtfully will score higher than a team that generated a long transcript with little reflection. Include the word count at the end of the document.

4 Milestones and Deadlines

4.1 Week 05: Proposal Discussion

During the Week 05 lab session, each team will lead a 15-minute class discussion on their chosen example of data visualization from news media. The discussion should address the following questions:

- What data are visualized?
- What does the visualization aim to communicate?
- What aspects of the visualization do you appreciate?
- What aspects do you find problematic?
- How might the visualization be improved?
- Who might be the target audience for an improved version, and what task or decision would the visualization support for that audience?
- Are the data publicly available, or are there suitable surrogate data?

The discussion is purely formative and will not be marked, but teams should incorporate feedback from classmates and instructors into their project delivery. The discussion is intended to be conversational; please do not prepare a slide show.

In preparation for the discussion, complete the following xSITe tasks by **Monday, 1 June 2026, 12:00 noon**:

- A discussion-list team entry (team ID, your chosen visualization from news media, and the URL to the original article) in the "Team Project: Thought-Provoking Data Visualization from News Media" discussion list.
- A short proposal note (up to 1,000 words) in the "Week 05 Formative Proposal Notes for Poster Project Discussion" xSITe Dropbox folder.

If the instructors deem the proposed project unsuitable, the team will be asked to revise its choice. If a suitable revision cannot be agreed on promptly, the instructors may assign an alternative visualization.

4.2 Week 09: Mid-Project Data Check-In

Before the Week 09 lab, each team must submit a brief Mid-Project Data Check-In to xSITe consisting of:

- The dataset the team has selected, in its raw and cleaned forms, together with a short description of its source.
- A QMD file containing the data import and cleaning pipeline.
- A draft of the design brief.
- A 1–2 paragraph note on the team's planned design direction and any questions to the instructors.

Submit the mid-project check-in to xSITe by **Monday, 29 June 2026, 09:00 AM**.

The check-in is marked solely on the basis of completeness. Its purpose is to surface problems early enough to correct them before the Week 11 consultation.

4.3 Week 11: Consultation Session

The instructors will be available for consultation during the lab session in Week 11. Teams should bring a current draft of the visualization and poster, share progress and challenges, and use the session to refine their design and presentation plan before the Week 13 lab.

4.4 Week 13: Poster Presentation

During the lab session in Week 13, teams will present their posters to their classmates and instructors. Students from other teams are asked to participate in the presentations and submit feedback to the instructors through a survey. Instructors will evaluate each team's presentation based on both class feedback and their own assessment.

5 Deliverables

The following items should be compressed into a single ZIP file and uploaded to xSITe by **Sunday, 26 July 2026, 23:59**.

- **Poster (PPTX file):** A PowerPoint file containing the poster in **16:9 aspect ratio**.
- **Original visualization:** A screenshot, PDF clipping, or image file showing the original published visualization, together with a citation including the source URL.
- **Data:** The raw data used for constructing the improved visualization.
- **Project QMD file:** This file should reference the data source and provide the complete pipeline for cleaning, transforming, and visualizing the data.
- **Rendered project report:** The HTML or PDF output produced from the project QMD file.
- **AI Interaction Record:** Raw, unedited transcripts of the conversations cited in the AI Usage Report, with the tool's name and version in each file name (see Section 3). No Record file is required for inline-completion tools such as GitHub Copilot.
- **AI Usage Report:** A QMD, Markdown, or DOCX file of ≤ 2,000 words, reflecting on the team's use of AI (see Section 3). Include the word count at the end of the document.
- **Additional materials:** Any supplementary materials necessary to render the QMD files.

6 Assessment

The project will be assessed based on the following criteria. The weights below are percentages of the total poster-project mark, which contributes 30% to your overall course grade:

Criterion	Description	Weight (%)
Mid-Project Data Check-In (Week 09)	• The dataset is provided in its raw and cleaned forms, along with a short description of its source. • A QMD file containing the data import and cleaning pipeline is included. • A draft of the design brief is included.	5

Criterion	Description	Weight (%)
	• A note on the planned design direction and any questions or blockers is included.	
Project motivation	• Information about the content, background, and significance of the data is communicated effectively. • The strengths and weaknesses of the original visualization are critically assessed. • Suggested changes to the original plot constitute improvements. • A suitable data source for the project work is presented, along with a succinct explanation of the raw data. • The project poses technical challenges commensurate with the course material. • The design brief identifies a specific audience, an analytical task, and a convincing rationale for the design choices.	10
Code quality	• The complete data transformation and visualization pipeline is included in the submitted files. • All auxiliary data files are included in the project submission to enable rendering the QMD file. • The code is free of logical errors. • The code is readable, consistently formatted, and documented. • Instructors can render the QMD file from the submitted ZIP archive without any edits.	10
Data visualization produced by the team	• The plot type is suitable for the data. • Any necessary data and coordinate transformations have been implemented and documented step-by-step. • Information in the plot is presented clearly and can be easily grasped. • The plot supports the written conclusions. • Legends, labels, titles, captions, and annotations are provided where necessary without cluttering the plot. • The solution showcases technical mastery and a creative application of the tools learned in this course. • The plot is visually polished, readable, and accessible. • The plot implements the design brief.	25
Poster and Design Brief	• Poster content is accurate and citations are correct. • Conclusions are correctly drawn from the data. • The steps from raw data to improved visualization are sequenced logically and concisely.	20

Criterion	Description	Weight (%)
	• The flow of information is coherent and logical. • Different sections are delineated with explicit headings and whitespace (e.g., introduction, design brief, original visualization, improved visualization, data). • The design brief is compact (roughly 50–125 words) and prominently placed; it names a specific audience and analytical task and links specific design choices to that task. • The quality of written English is high. • The visual layout of the poster is appealing and easy to navigate. • Font size and style are appropriate and legible from a distance.	
Oral presentation	• Speakers maintain eye contact with the audience and refrain from reading text from their mobile phones. • The presentation is well-rehearsed, with all team members contributing effectively. • The significance of the data and the project is stated explicitly. • The design brief is presented and motivated. • The quality of spoken English is high. • The time limit is adhered to. • Questions from the audience are answered competently and effectively.	10
AI usage and reflection	• Specific claims made in the AI Usage Report are substantiated by the corresponding raw transcripts in the AI Interaction Record. • Prompting strategies are described clearly, with evidence of the team's reflection. • At least two specific design or implementation decisions are traced to AI input, with explicit identification of what was accepted, refined, or rejected. • Critical engagement is evident: the team identifies cases where AI was wrong, biased, or misleading and explains how they corrected for it. • The team's engagement with AI is ethical and accountable, including appropriate attribution and verification of factual or technical claims.	15
Peer feedback and attendance	• All teammates must attend their allotted lab sessions (unless excused).	5

Criterion	Description	Weight (%)
	• Students provide constructive feedback on other teams' posters and presentations through the required survey.	